

Seonghyun Park

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Summary

I'm a second-year Ph.D. student at [Graduate School of AI at KAIST](#), advised by [Sungsoo Ahn](#). My research interests lie mainly in **AI4Science**, and I believe that machine learning can advance scientific discovery involving small molecules and proteins. Recently, I have been focusing on incorporating experimental data into machine learning models to better understand protein-ligand interactions.

Education

Ph.D. Korea Advanced Institute of Science and Technology (KAIST) , Kim Jaechul Graduate School of Artificial Intelligence • Structured and Probabilistic Machine Learning Lab @ Sungsoo Ahn • Interest: Protein-drug interactions, Molecular dynamics (MD)	Seoul, South Korea Feb 2025 – present
M.S. Pohang University of Science and Technology (POSTECH) , Graduate School of Computer Science and Engineering (CSE) • Machine Learning Lab @ Sungsoo Ahn • Interest: Graph Neural Networks (GNNs), Over-squashing	Pohang, South Korea Mar 2023 – Feb 2025
E.S. Institut National des Sciences Appliquées (INSA) Lyon , Bioinformatics Exchange Student	Lyon, France Jan 2022 – June 2022
B.S. Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering (CSE)	Pohang, South Korea Feb 2019 – Feb 2023

Experience

Bagelcode , Business Analyst (BA) intern Game economy management and KPI analysis automation	Seoul, South Korea June 2021 – Aug 2021 3 months
Seller Hub , Product Manager (PM) intern Organization-wide task prioritization and landing funnel renewal	Seoul, South Korea July 2020 – Aug 2020 2 months

Publication

† Equal contribution · ‡ Co-corresponding author

Blending experimental electron density information to protein-ligand interactions

Seonghyun Park, Shuan Chen, Jun Jin Choong, Noboru Kuno, Gyubok Lee, Kaushalya Madhawa, Keisuke Ozawa, Hikaru Shindo, Sungsoo Ahn

A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning

Hyosoon Jang, Hyunjin Seo, Honghui Kim, **Seonghyun Park**, Taewon Kim, Yunhui Jang, Sungsoo Ahn
 NeurIPS 2026 ([10.48550/arXiv.2602.13249](#))

INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery

Yunhui Jang, **Seonghyun Park**, Jaehyung Kim, Sungsoo Ahn
 ICML 2026 Workshop, AI4Science ([10.48550/arXiv.2602.01815](#))

Riemannian MeanFlow

Dongyeop Woo, Marta Skreta, **Seonghyun Park**, Kirill Neklyudov, Sungsoo Ahn
ICML 2026 ([10.48550/arXiv.2602.07744](https://arxiv.org/abs/10.48550/arXiv.2602.07744))

Learning Collective Variables from BioEmu with Time-lagged Generation

Seonghyun Park, Kiyoun Seong, Soojung Yang, Rafael Gomez-Bombarelli, Sungsoo Ahn
ICLR 2026 ([10.48550/arXiv.2507.07390](https://arxiv.org/abs/10.48550/arXiv.2507.07390))

Transition Path Sampling with Improved Off-Policy Training of Diffusion Path Samplers

Kiyoun Seong, **Seonghyun Park**, Seonghwan Kim, Woo Youn Kim, Sungsoo Ahn
ICLR 2025 ([10.48550/arXiv.2405.19961](https://arxiv.org/abs/10.48550/arXiv.2405.19961))

Non-backtracking Graph Neural Networks

Seonghyun Park[†], Narae Ryu[†], Gahee Kim, Dongyeop Woo, Se-Young Yun[‡], Sungsoo Ahn[‡]
TMLR 2024, NeurIPS 2023 Workshop GLFrontiers Oral ([10.48550/arXiv.2310.07430](https://arxiv.org/abs/10.48550/arXiv.2310.07430))

Diffusion Probabilistic Models for Structured Node Classification

Hyosoon Jang, **Seonghyun Park**, Sangwoo Mo, Sungsoo Ahn
NeurIPS 2023 ([10.48550/arXiv.2302.10506](https://arxiv.org/abs/10.48550/arXiv.2302.10506))